

IMCQ 9 DM

KEY

Q#1a: A 34-year-old man takes antacids on a regular basis to treat his heartburn that is caused by acid reflux. His physician suggests that he should reduce the usage of these drugs as they interfere with his normal digestion. Which of the following processes needs a high proton concentration?



- A. Activation of phospholipase A_2
- B. Activation of salivary amylase
- C. Activation of pepsinogen
- D. Activation of enteropeptidase
- E. Activation of trypsinogen

Q#2: A 3-year-old infant is treated for a severe lung infection. She is undernourished and her sweat is very salty. Her treatment includes a specific diet which improves her deficient energy metabolism. Which of the following nutrients is most likely used to enrich her diet?



- A. TAGs with eicosapentaenoic acid
- B. Proteins found in whey
- C. TAGs with medium-chain fatty acids
- D. Proteins found in casein
- E. TAGs with dietary essential fatty acids

Q#3: A 46-year-old woman has severe indigestion resulting from a deficient release of cholecystokinin. Which of the following actions are directly regulated and stimulated by this hormone?



- A. Release of trypsin from the pancreas
- B. Release of bile from the gallbladder
- C. Release of hepatic bile salts into bile
- D. Release of chyme into the duodenum
- E. Release of chylomicrons into the lymph

Q#4: A research group investigates the dietary uptake of nutrients into intestinal mucosal cells. In their experiments they inhibit sodium-potassium ATPase and compare their data with the uninhibited conditions. Which of the following nutrients is taken up in spite of the inhibited enzyme?



- A. All amino acids
- B. Dietary essential amino acids
- C. Glucose
- D. Fructose
- E. Galactose

Q#5: A research group analyzes the conformational change that follows when trypsinogen is activated to trypsin. They find that a long narrow active site is formed. Which of the following amino acids has a side chain that is locked into this pocket by ionic bond when the protein is cleaved by trypsin?



- A. Lysine
- B. Histidine
- C. Asparagine
- D. Tyrosine
- E. Aspartate

Q#6: A 54-year-old man experiences severe pain in the abdomen and is brought to the emergency room. Two serum enzyme markers for pancreatitis are elevated, one is amylase. What is characteristic for the normal function of the other elevated serum enzyme?



- A. The enzyme needs bile salts for its activity
- B. The enzyme degrades TAGs to free glycerol
- C. The enzyme degrades proteins
- D. The enzyme needs high proton concentration
- E. The enzyme needs activation by enteropeptidase

Q#7: A 58-year-old male is treated in the hospital for hypoglycemia and lactic acidemia. He is a chronic alcoholic who did not eat a meal but instead ingested large amounts of alcohol the night before. Which of the following hepatic pathways is reduced due to his abnormal high NADH/NAD⁺ ratio in cytosol and mitochondria and leads to his symptoms?



- A. Urea cycle
- B. Bile acid synthesis
- C. Gluconeogenesis
- D. Ketone body synthesis
- E. Cholesterol synthesis

Q#8: A 58-year-old man is treated in the hospital. He consumed excessive ethanol for several years which lead to his liver disease. Which of the following molecules directly damages hepatic tubulins when it accumulates in cytosol?



- A. Acetate
- B. Ketone bodies
- C. Ethanol
- D. Acetaldehyde
- E. TAGs

Q#9: At highly elevated ethanol levels, hepatocytes use ethanol as substrate for both alcohol dehydrogenase and for MEOS. What is characteristic for MEOS?



- A. It uses NADH as coenzyme
- B. It contains mainly CYP2E1
- C. It scavenges radicals
- D. It forms acetate
- E. It is found in mitochondria

Q#10: A 25-year-old healthy woman participates in research that analyzes her metabolic homeostasis during fasting. What is characteristic for her blood levels found on day 3 of this study ?



- A. Blood glucose level is back to 5 mM
- B. β -hydroxybutyrate level increases sharply
- C. Free fatty acid level drops nearly to zero
- D. Blood glucose level drops to 2.8 mM
- E. Acetoacetate level is higher than β -hydroxybutyrate

KEY

1. C

2. C

3. B

4. D

5. A

6. A

7. C

8. D

9. B

10. B